explaining the confusing structure of the Standard Model and its parameters. It came with a tantalizing promise too: a whole slew of new particles ("supersymmetric partners" of the Standard Model set) that might be produced in particle collisions just a *little* more powerful than what could be achieved by colliders at that time. SUSY has also been widely held up as a stepping-stone toward string theory, the leading idea in the quest to bring gravity and quantum mechanics together into a unified whole.

Unfortunately, despite working for decades to improve and upgrade the LHC, we've seen no sign of supersymmetry's promised particles. Some physicists still hold out hope for SUSY by proposing tweaks that would make the new particles harder to find, but at some point the tweaks become so extreme that SUSY has just as many theoretical problems as the Standard Model. And the signal just isn't there. Now and then, some quirk of the data will produce a whirlwind of excitement as physicists rush to explain why there are a few more events in a particular detector channel than expected. But so far, none of these blips has turned out to be more than a statistical fluke destined to fade away in the next data release.

I spoke to Freya Blekman, an experimental physicist who searches for beyond–Standard Model signatures in LHC data, about the current conundrum. "I've been in the field twenty years now and I've seen my share of excesses come and go, and I've also seen my share of popular models come and go," she said. "Depending on who you speak to, there are people who are disillusioned … people have been telling them for a very long time that they should be seeing something. And what the experiments see is only the Standard Model." From her perspective, though, the disillusionment is misplaced. Not because people are missing hints that are really there after all, but just because there was never any guarantee that anything new would be found with these experiments.

Still, the lack of direction from experiments can be troubling—enough to push some researchers out of particle physics entirely, and into cosmology. One of those is Pedro Ferreira, a cosmologist at Oxford University who switched from quantum gravity to cosmology during his PhD and who now studies the cosmic microwave background and general relativity in astrophysics in the hope that they might provide some better insights. "There hasn't been anything revolutionary that particle theory has done which has led to observational results since 1973," he says. There have been lots of new theoretical ideas, and some of them are very appealing, but